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REGULAR PAPER

Trends, prospects, challenges, and security in the healthcare internet of things

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Abstract

The Healthcare Internet of Things (H-IoT) is a rapidly developing problem solving model with significant potential to improve patient care and healthcare outcomes. This study focuses on integrating cryptographic platforms into H-IoT systems to enable secure data access. We present insights on how to manage challenges such as cyber risks, resource constraints, latency, and energy consumption by exploring cryptographic approaches in diverse H-IoT applications. We explore important topics including big data management, blockchain, machine learning, edge computing, and software-defined networks. Additionally, we examine real-time operational challenges and emerging trends, such as remote patient monitoring and predictive analytics. Our research underscores the need for strong encryption mechanisms, access controls, device authentication, and proactive threat detection to improve the safety and efficiency of healthcare services. By combining cryptographic approaches with the architecture of the H-IoT system, this work provides the foundation for resilient healthcare infrastructures. Addressing current challenges and anticipating future opportunities contribute to strategic healthcare planning that prioritizes patient privacy and data security while anticipating future strategic healthcare planning opportunities. This article aims to provide valuable information to researchers by covering energy-efficient and resource-optimized healthcare system approaches that are integrated into the development of H-IoT systems.

Keywords Healthcare · IoT · Blockchain · Edge computing · SDN · ML · DL

Mathematics Subject Classification 94A60 · 68M12 · 68U35 · 68T01 · 68T07 · 68T30

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1 Introduction

Numerous healthcare apps have been developed in the last few decades with the goal of improving medical procedures. A new era of electronic healthcare and industrial research has arrived as a result of recent trends and advances in information technology [1]. The Internet of Things (IoT) is a confluence of several communication technologies that can sense and communicate with one another. By resolving historical difficulties and complications, IoT devices have upended established procedures in the field of biomedical applications [2]. These Internet of Things (IoT) devices have the potential to provide large volumes of biological data, which can therefore be used to create automated medical data collection systems [3]. Big data plays a crucial role in improving healthcare systems, notably in the areas of diagnosis, treatment, and decision-making, when IoT devices are combined with cutting-edge Machine Learning (ML) algorithms. The Internet of Everything (IoE) as a whole has attracted substantial academic attention thanks to biomedical IoT applications, which include things such as symptomatic therapies and patient monitoring [4]. Additionally, the introduction of tiny sensors into the medical field has made it possible to continuously monitor patients' vital signs, strengthening the security of the human healthcare system and increasing the possibility of early diagnosis and treatment [5].

Healthcare professionals now have access to a large database of patient health data due to the widespread use of wearable technology, sensors, and other linked devices [6]. Using this information will improve patient outcomes and raise the standard of treatment. However, healthcare providers must establish efficient information processing systems in order to fully utilize these data. The collection, storage, analysis, and distribution of data produced by IoT devices are all included in information processing in the Healthcare Internet of Things (H-IoT). Important parameters such as blood pressure, heart rate, oxygen saturation, patient activity levels, sleep patterns, and medication adherence are all included in this data. With these data at their disposal, healthcare professionals can identify early symptoms of disease, observe patients remotely, and customize treatment approaches, thus improving healthcare outcomes [7]. Data gathering is the first step in information processing in the HIoT and is accomplished by a network of wearables, sensors, and other associated devices. Afterwards, these data are sent to a central database or cloud-based storage system. The architecture of this database is crucial because it must be able to manage large amounts of data while ensuring accessibility for healthcare practitioners in various places. Data analysis comes next once the data has been gathered and safely stored. This involves using sophisticated algorithms to find patterns and trends in the data, which could include identifying changes in vital signs, detecting irregularities in patient behavior, and foretelling probable future health problems. Data dissemination is the last stage in the processing of H-IoT information. Healthcare providers need to get the data in a manner that is both simple to use and understandable. This sometimes entails highlighting important trends and patterns in the data using visual tools such as graphs and charts. Additionally, alarm systems and alerts may be set up to immediately alert medical professionals to significant alterations in a patient's health state.

Healthcare providers are facing a special set of issues as a result of the expanding usage of H-IoT devices, which produce large amounts of data and might eventually become too much for them [8]. The mind-sponge idea has been proposed as a solution to these problems. This idea states that healthcare professionals must learn to "sponge up" these data and analyze them in a way that allows informed decision-making and the delivery of improved patient care. Advanced analytics and machine learning algorithms are used to uncover data patterns and trends and prioritize the most important information for physicians in order to achieve this goal. Healthcare practitioners may greatly improve patient outcomes and overall care quality by utilizing the potential of IoT data in this way [9, 10]. The proliferation of IoT systems has caused significant security problems, particularly due to the quick creation and spread of these devices. Given that data breaches in the healthcare Internet of Things (HIoT) can have serious consequences, including the possible loss of life, security concerns are of special importance. The revolutionary potential of H-IoT in a hospital setting is depicted in Fig. 1. For instance, people with diabetes have ID cards that may be scanned and connected to a cloud-based network. Electronic Health data (EHRs), prescriptions, significant test results, and medical history data are all stored on this platform. This arrangement simplifies access for nurses and physicians, who may easily access these documents using PC, laptop, or tablet computers.

Although it is of the utmost significance to guarantee people's comfort and security when disclosing their private health information, doing so is nevertheless

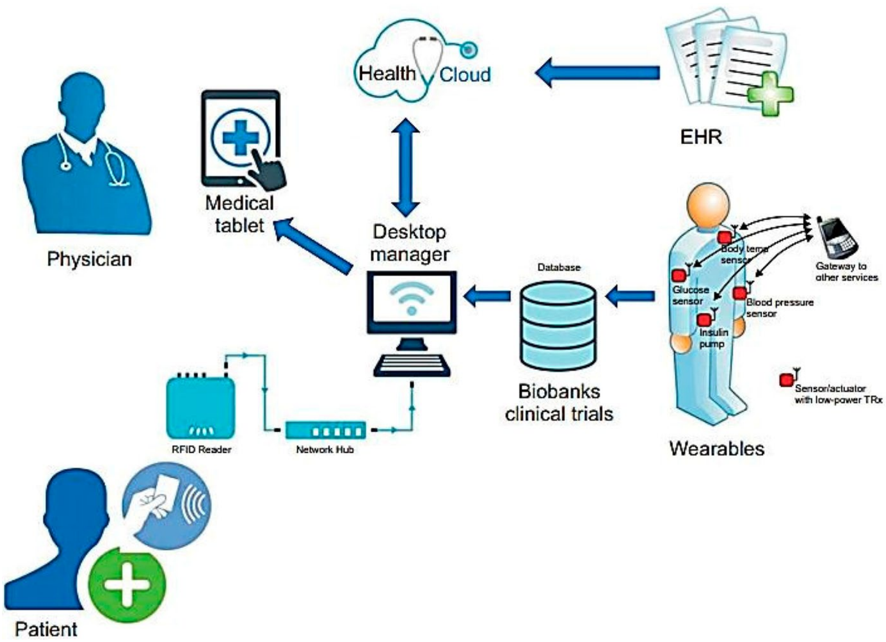

Fig. 1 Revolutionary aspects of HIoT in a medical setting

difficult as a result of the many security rules and access control systems in place. Maintaining both privacy and the integrity of medical data requires careful balancing of both factors, which is a challenging challenge. Several cryptographic platforms, including blockchain, big data, software-defined networks, edge computing, and artificial intelligence, have emerged in response to security problems in the (HIoT). These technologies have been developing in an exciting new direction with IoT advances in the healthcare industry. By improving data security, efficiency, and management precision, as well as by providing better diagnostic and treatment choices, the integration of these platforms into H-IoT has the potential to transform the healthcare industry. However, it is crucial to recognize that the deployment of these technologies may be complex, expensive, and frequently requires specific expertise. Managing and interpreting large data quantities may be difficult, and laws governing data ownership and responsibility are continuously developing. Additionally, specialized hardware and software infrastructure would be required, and not all H-IoT applications are appropriate for edge computing. The expense of creating and maintaining the necessary infrastructure is a major issue with H-IoT. The integration of IoT devices, sensors, and wearables into current healthcare systems could also bring financial challenges due to their high cost. To manage the data produced by IoT devices, healthcare providers may need to spend money on specialized employees, such as data analysts and IT specialists. Additionally, recurring maintenance fees, which include charges for things like battery replacements, software updates, and gadget replacements, can be high. These expenses may be especially onerous for smaller healthcare facilities or those working with limited resources. Therefore, before making significant investments in H-IoT, it is essential to perform a complete cost-benefit analysis of this technology.

The issue of 'near suicide', in which people forgo medical care due to financial worries and suffer lethal consequences, was studied by Vuong et al. [11]. Seriously sick individuals were more likely to discontinue treatment if the cost would have a major negative influence on their family's financial situation, according to their analysis of a dataset of 1042 Vietnamese patients using BMF analytics. This shows how crucial it is to take into account the financial effects of healthcare technology when designing real-world healthcare systems. With current trends focusing on wearable technology, predictive analytics, remote patient monitoring, and the development of intelligent hospitals, the HIoT industry is expanding quickly and dynamically. Artificial intelligence (AI) and telemedicine together are expected to result in more accurate diagnoses, greater patient participation, and better healthcare results. Future-looking potential for improved HIoT systems includes using health data for strategic planning, customizing treatment plans to meet the requirements of specific patients, and combining cutting-edge medical equipment with IoT technology. However, in order to ensure the secure and efficient usage of H-IoT, it is crucial to solve issues related to data privacy and security. High fault tolerance, strong data security, dependable Quality of Service (QoS) criteria (such as low power consumption), minimal latency, data integrity scalability, and interoperability are just a few of the key characteristics of a reliable IoT system in the healthcare industry. This article also examines the aspects of H-IoT systems' mobility, real-time data processing, and deployment simplicity.

Despite the increasing data about the spread of HIoT systems, current studies frequently fall short of thoroughly addressing all important features of H-IoT. This could lead to the absence of important advancements, technologies, and applications, which would result in a lack of understanding of the subject. To provide a more comprehensive and informative analysis, it must be crucial to fill up these gaps and encourage more study. Future scholars will greatly benefit from the innovative ideas and concepts introduced in this study, which bring a fresh and insightful viewpoint to the area.

The organization of the paper is as follows:. HIoT is covered in detail in Sect. 2, including H-IoT in edge computing, H-IoT in big data, and H-IoT in machine learning and deep learning. In Sect. 3, Interdisciplinary Viewpoints on HIoT Adoption in Healthcare is presented, and Sect. 4 presented the practical challenges in HIoT deployment. In Sect. 5, we examine the meaning of HIoT in the medical sector, and in Sect. 6, we conduct a HIoT applications. Furthermore, Sect. 7 explores the critical analysis. Section 8 serves as the paper's conclusion. Finally in Sect. 9 present the Future Research Directions as the last section.

2 Healthcare internet of things

H-IoT is an area that is quickly developing in the healthcare industry that integrates internet-connected devices, sensors, and software applications [12, 13]. Its main goal is to provide real-time data, predictive analytics, and remote monitoring to improve patient outcomes, increase efficiency, and save costs. Wearable technology, remote patient monitoring, telemedicine, and intelligent hospital systems are all examples of H-IoT applications [14]. By providing individualized and effective treatment, the HIoT has the potential to revolutionize healthcare delivery through the integration of artificial intelligence and machine learning. However, to ensure the secure and efficient application of HIoT technologies, addressing data confidentiality and security is a crucial barrier that must be overcome. IoT has recently been regarded as one of the healthcare industry's most promising technologies. In this regard, Rodrigues et al. [15] have presented a detailed analysis of IoT in healthcare, also known as HIoT. This research also outlines developments in HIoT technology as well as any connected problems that need to be fixed. To solve the existing issues facing HIoT, Rodrigues et al. have also stressed the need for more study in this area. Tsafack et al. have proposed a strong cryptographic solution for the safe transmission of MRI images within the framework of the Healthcare Internet of Things (H-IoT) [16]. The complexity of a 2D trigonometric map, which has infinite solutions, was studied in depth. The complex dynamics of this map were further shown in the paper using phase portraits, bifurcation diagrams, and the Lyapunov exponent. As a potential candidate for inclusion in HIoT to ensure safe transmission of medical pictures, the performance evaluation of the cryptosystem suggested in the study strongly suggests that it is highly secure. Figure 2 demonstrates how IoT is being more widely accepted and used in the healthcare industry with the aim of improving patient outcomes, increasing efficiency, and lowering costs.

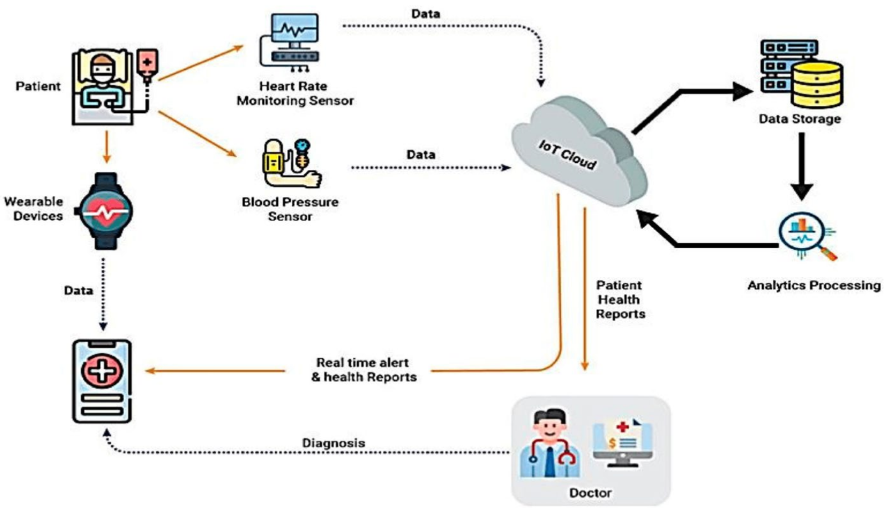

Fig. 2 Revolutionary features of H-IoT in a hospital environment

2.1 HIoT in bigdata

In order to gather and analyze enormous volumes of medical data, the Healthcare Internet of Things (HIoT) is significantly dependent on big data. Predictive models that can help identify possible health hazards and forecast the progression of diseases have been developed as a result of the integration of big data into the H-IoT [17]. In order to develop individualized treatment regimens for individuals, healthcare professionals might examine large databases using machine learning algorithms. Taking these factors into account, big data analytics inside the HIoT can elucidate patterns and trends, ultimately resulting in improved patient outcomes, lower healthcare costs, and more effective healthcare delivery. e-health services can be provided by a customized healthcare system to meet the clinical requirements of the elderly population. Jagadeeswari et al. [18] carried out a thorough investigation of new technologies in the healthcare sector, including big data analytics, mobile apps, the Internet of Things, fog computing and cloud computing, was carried out by Jagadeeswari et al. [18]. This study also examined the difficulties in developing a better health system for early detection of diseases and offered suggestions for feasible ways to get beyond these restrictions. Our evaluation aims to establish the foundation for the future creation of a better healthcare system.

The Healthcare Internet of Things (H-IoT) has an intelligent system that Yang et al. [19] have proposed that not only manages huge data storage, but also ensures data security. Adaptive access control is a key component of the system that is being developed, providing strong security for patient health information. In order to save storage space for large datasets, it offers effective data de-duplication and distinguishes between access control needs for emergency and normal scenarios. A system created for safe exchange among healthcare professionals working in various medical sectors houses the encrypted medical records generated by the H-IoT.

Furthermore, these access control systems let authorized users decrypt private medical data, permitting quick medical assistance in life-threatening circumstances. The research [20] proposes a brand new safe access management system that can be used in emergency and regular situations. Using a "break glass" access method, patient medical data may be retrieved in urgent situations. The study also validates the safety and efficiency of the suggested framework. Big data is essential for many business applications, including robots and healthcare Internet of Things (HIoT) technologies. Big data analytics are heavily dependent on H-IoT-driven systems. However, when working with sensory data, making precise predictions may be difficult. In order to solve this problem, Sivaparthipan et al. [21] devised an HIoT system that uses artificial intelligence (AI) to cure Parkinson's disease (PD) and improve the gait of affected people. This research also briefly highlights the important role that robots play in the management of Parkinson's disease. Robotic PD treatment strategies include anticipating walker movements and providing physical therapy. Big data exchanges are also facilitated by robotics.

Over the past ten years, the use of big data in the healthcare industry has grown rapidly, inspiring the publication of several cutting-edge findings and procedures. Big data approaches have been shown to be useful tools in biomedical informatics and healthcare research. Researchers, hospitals, patients, sensors, and mobile phones have recently come together to create a substantial amount of healthcare data. Clinical organizations continuously generate these data with the aim of identifying and treating new illnesses. Although patients have started using mobile applications to manage their recovery requirements, researchers have also created a variety of methods for collecting data. Additionally, the Internet of Things (IoT) enables seamless integration of these gadgets and apps with telehealth and telemedicine. Mishra et al. [22] looked at the problems related to big data's usage in the biomedical field. Through the incorporation of big data technology and distributed computing systems, they presented novel techniques to enhance healthcare systems. As shown in Fig. 3, IoT is considered a key development in the age of big data, since it offers a variety of real-time applications and improved services.

The Internet of Things (IoT) often requires that each device produce large amounts of data, necessitating effective processing and storage systems. The

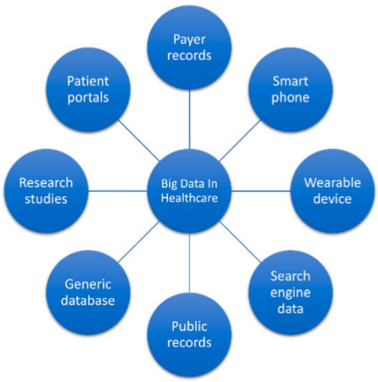

Fig. 3 Big data applications in the healthcare sector [23]

emergence of cloud computing has provided a remedy for this issue. However, real-time data processing is essential for healthcare applications to improve performance and reduce latency. In the healthcare industry, fog computing stands out as a possible option because it can reduce data connection delays, provide flexible services, and more efficiently distribute resource needs. A fog-based healthcare Internet of Things (HIoT) system was also suggested by Feng et al. [24] with the purpose of lowering energy usage and network latency. The report made recommendations and provided an overview of crucial services of big data infrastructure for fog devices that analyze medical data. Numerous areas of life, including communication, transportation, healthcare, surveillance, business, and a variety of other sectors, find use for smart sensors and IoT devices. These sensors and gadgets provide enormous volumes of data, which, when further analyzed, may be quite important for healthcare companies [25].

2.2 HIoT in Edge_Computing

Generally speaking, the Internet of Things (IoT) links many smart sensing devices over numerous networks, and these devices are incorporated into a variety of applications, including smart grids, smart homes, and smart healthcare systems [26–28]. The security of IoT devices and data is crucial when thinking about IoT-based healthcare solutions [29]. In the IoT, edge computing is developing as a potential component that addresses complicated processing problems and aims to deliver lower-latency data by accelerating IoT device computation and communication [30]. In edge healthcare systems based on the Internet of Things, load balancing and effective resource utilization may be encouraged by the use of AI and network optimization [31]. It's crucial to keep in mind that IoT devices frequently have limited power, making the data they store more vulnerable to security risks [32]. The creation of a safe framework for IoT-enabled edge computing-based healthcare systems using software-defined networks (SDN) is therefore essential [33, 34]. This strategy uses edge servers to authenticate IoT devices using a simple authentication process. These IoT devices capture and transfer patient data to edge servers for archival, processing, and analysis after authentication. An SDN controller links these edge servers together, enabling healthcare systems to better use resources, optimize their networks, and balance load. The results of the study, which examined the suggested methodology using computer-based simulations, showed that it offered better solutions for healthcare systems based on edge computing and IoT.

As shown in [35, 36], improvements in sensing methodologies have sparked the creation of intelligent systems capable of continuously monitoring human behavior. Recurrent Neural Networks (RNN) based wearable sensor-based frameworks for activity prediction on PCs and laptops have grown in popularity. The suggested system used input data from a variety of wearable healthcare sensors, including gyroscopes, accelerometers, magnetometers, and ECG sensors (electrocardiography). Using readily available datasets, the study compared the proposed strategy with conventional methodologies. The experimental findings proved that the proposed strategy outperforms the more established ones. The creation of efficient healthcare applications, such as remote patient monitoring via centralized electronic clouds (E-clouds), has been made easier by the introduction of edge computing. High-quality service (QoS), sophisticated sensing capacities, and crystal-clear vision are required for this, with the latter being given great importance. [37] established a window-based rate control method (W-RCA) to improve medical QoS (M-QoS) in edge computing to solve this problem. The peak-to-mean ratio (PMR), latency, standard deviation (SD), and network jitter are among the factors taken into account in this optimization. In order to improve MQoS, the study evaluated the proposed W-RCA method and contrasted it with other existing algorithms, such as the Battery Smoothing method (BSA). According to experimental research, the recommended approach works better than other ones that are already in use, especially for applications like telesurgery.

The usage of smart systems has been encouraged by the fast development of information technology and the Internet of Things (IoT). In addition, intelligent healthcare systems in developed and smart cities depend on cloud and edge computing. For modeling, simulating, and improving service flow and resource allocation in emergency departments (ED), real-time systems and customized techniques are needed. As a result, [38] provided theoretical justifications and validation for a method for modeling non-consumable resources. To account for performance metrics including average patient waiting times, resource utilization rates, and patient lengths of stay (LoS), the study used Realistic Performance Notation (RPN) in real-world settings. Improvements in LoS, patient wait times, and resource use were highlighted in the simulation of the proposed system.

2.3 HIoT for deep learning and machine learning

In the context of the healthcare Internet of Things (H-IoT), deep learning and machine learning techniques are being rapidly used [39, 40]. The authors [41, 42] describe a unique structure of deep learning (DL) based on HIoT that uses the idea of transfer learning to identify and categorize cervical cancer. Convolutional neural networks (CNN) were combined in the study with a number of well-known machine learning techniques, such as Support Vector Machines (SVM), Random Forests, Logistic Regressions (LR), Naive Bayes (NB), and K-Nearest Neighbors (KNN). CNN models pre-trained including ResNet50, SqueezeNet, VGG19, and Inception V3 were used for feature extraction. The performance of the H-IoT system was evaluated, and the results showed that the proposed model had a high classification rate of 97.89%. A DL technique was used in a different study [43–45] to effectively categorize and predict Chronic Kidney Disease (CKD). The model used a Softmax classifier with stacked auto-encoders (AEs). The dataset was mined for features and multi-modal patterns using AEs, which improved classification performance. Through feature reduction for individual AEs, this method facilitated classification while utilizing the Softmax classifier for sample classification. The Deep Stacked AE model demonstrated good accuracy, specificity, recall, and precision-key characteristics in the medical industry-and was shown to be a useful diagnostic tool for CKD. In addition, a DL algorithm was developed and tested to predict CKD from retinal images in [46]. In [47] a cluster CNN architecture was recommended for the diagnosis of P. vivax, which automatically improved the diagnosis of malaria without depending on the characteristics created manually. Higher parasite identification accuracy was attained with the use of fivefold cross-validation, proving that different convolutional layer depths and filter sizes may extract different characteristics for classification.

2.4 HIoT in SDN

Instead of using the conventional administrative paradigm, network nodes can now be managed using programming languages thanks to a software-defined approach to networking [48, 49]. The Internet of Things (IoT) uses software-defined networks (SDNs) to address a number of issues. By detaching physical device control, they offer solutions for network virtualization, network management, energy management, resource usage, privacy, and security [50]. Separation of the control plane from the data plane is one of the main purposes of SDN, which improves network performance [51]. Using a simple authentication mechanism, the authors in [52, 53] created a safe framework to authenticate IoT devices using edge servers. Following authentication, these devices collect patient information and send it to edge servers for archiving, processing, and analysis. These edge servers are connected to an SDN controller, which manages load balancing, network optimization, and effective resource use. Computer simulations were used to assess the suggested framework, with positive outcomes. It is crucial to guarantee the confidentiality of private medical records and reports kept in the cloud [54]. This technology quickly alerts medical personnel, facilitating the management of numerous disorders. An SDN-based structure with dedicated virtual machines for each patient in the data sharing system that can be made available to authorized users was suggested in [55, 56]. The virtual machine is also secured by an SDN-based gateway that serves as a firewall, guaranteeing patient privacy and access through virtual machines with distinctive MAC addresses.

2.5 HIoT in blockchain

Access control for healthcare systems may benefit from the distributed ledger technology known as blockchain, according to [57]. Although switching to a full blockchain solution might be difficult, several implementations and concepts have been investigated. In [58] writers put forth a unique blockchain-based system that used swarm exchange techniques to make it easier to send electronic health information securely and quickly. They included announcement, swarm intelligence, peer open, and peer shutting algorithms to boost the widespread Electronic Health Record (EHR) transmission for better healthcare services. The system, which used IoT health sensor nodes to track vital indications such as heart rate, body temperature, and oxygen saturation, was created using tools like IPFS, GnuPG, and Golang. In [59], IoT sensors and blockchain smart contracts were used to monitor the structural health of subterranean constructions. To improve operational safety, our strategy built a distributed network that is secure and scalable. For scalability and efficiency, the system made use of locally and globally centralized distribution, classifying them into edge and core networks. Using this method, controlled structures could be simulated and autonomous monitoring was made possible. The authors in [60, 61] investigated using blockchain-based smart contracts to monitor patient vital signs. They used Hyperledger Fabric to construct the system, which offered advantages to everyone, regardless of where they were located. Physiological information was collected using the Linoleum Toolkit. Using common benchmarks with Hyperledger Caliper, the framework's efficiency was assessed in terms of latency, resource use, and transactions per second. Compared to conventional healthcare systems, the findings demonstrated that this system effectively tracked patient data. The writers suggested a blockchain-based system that anticipated consumer status and paid them according to established standards in [62]. The results of the model showed that this customer-centric strategy aided in corporate transformation.

3 Interdisciplinary viewpoints on HIoT adoption in healthcare

The adoption of Healthcare Internet of Things (HIoT) technologies is a complex process that benefits significantly from the insights and collaboration of various disciplines. Leveraging expertise from computer science, healthcare professionals, healthcare management, and policymakers is essential to gain a comprehensive understanding of the challenges and opportunities associated with HIoT [63]. Figure 4 illustrates the interdisciplinary viewpoints on HIoT adoption in healthcare.

3.1 Computer science

Computer scientists play a crucial role in the development and implementation of HIoT technologies. Their expertise is essential for addressing several key challenges and opportunities in this field.

3.1.1 Challenges

One of the primary challenges is ensuring the interoperability of diverse HIoT devices and systems. Standardized protocols and interfaces are required to enable seamless communication between different devices. Additionally, the vast amounts of data generated by HIoT devices necessitate advanced data processing techniques. Cybersecurity is another significant challenge, as the protection of sensitive health information is paramount.

3.1.2 Opportunities

Advancements in machine learning and big data analytics present opportunities to enhance the functionality of HIoT systems. These technologies can be used to analyze large datasets, providing valuable insights for improving patient care

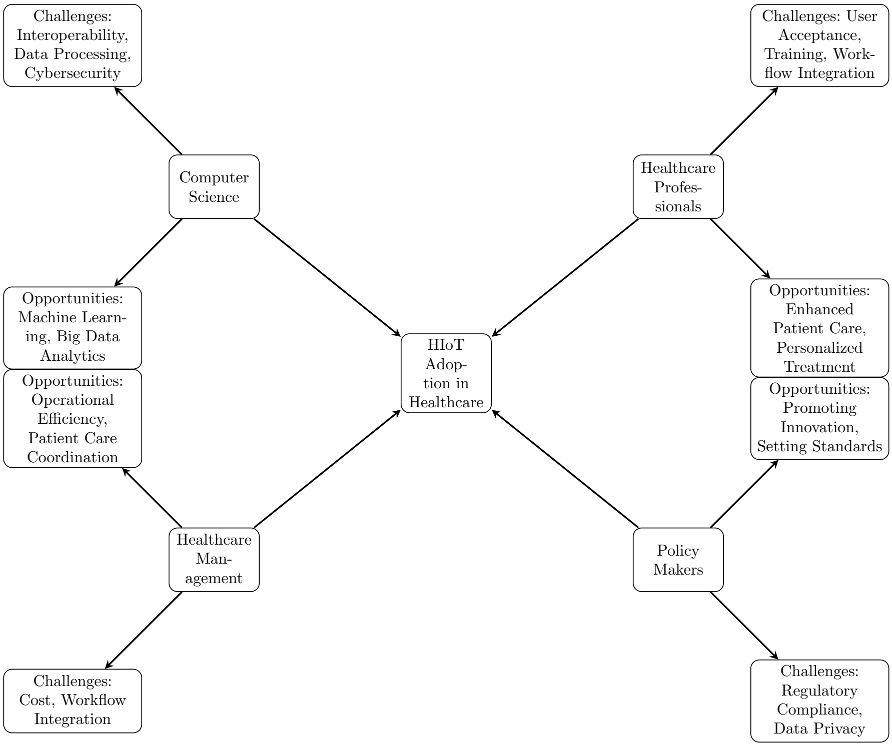

Fig. 4 Interdisciplinary Viewpoints on HIoT Adoption in Healthcare

and operational efficiency. Furthermore, the development of robust cybersecurity measures can help protect patient data, fostering trust in HIoT systems.

3.2 Healthcare professionals

Healthcare professionals, including doctors, nurses, and medical researchers, are integral to the successful adoption of HIoT technologies. Their insights and experiences are vital for addressing practical challenges and leveraging opportunities.

3.2.1 Challenges

User acceptance is a significant challenge in the adoption of HIoT technologies. Healthcare professionals must be willing to incorporate these devices into their clinical practice. Training and education are crucial for increasing user acceptance and ensuring that healthcare professionals are proficient in using HIoT devices. Additionally, integrating HIoT data into clinical workflows can be challenging.

3.2.2 Opportunities

HIoT technologies offer numerous opportunities to enhance patient care. Continuous monitoring allows for early detection of health issues, leading to timely interventions and improved patient outcomes. Personalized treatment plans can be developed based on the data collected from HIoT devices, ensuring that patients receive care tailored to their specific needs.

3.3 Healthcare management

Healthcare administrators play a pivotal role in implementing HIoT systems within healthcare facilities. Their expertise is essential for managing the financial and operational aspects of HIoT adoption.

3.3.1 Challenges

The cost of HIoT devices and infrastructure can be a significant barrier to adoption, particularly for smaller healthcare facilities. Effective cost management strategies are needed to ensure that the benefits of HIoT outweigh the initial investment. Additionally, integrating HIoT systems into existing workflows can be complex, requiring careful planning and execution.

3.3.2 Opportunities

HIoT technologies have the potential to improve operational efficiency in healthcare settings. Automated data collection and analysis can streamline administrative tasks, reduce manual errors, and optimize resource utilization. Improved coordination of patient care can also be achieved through better data sharing and communication facilitated by HIoT systems.

3.4 Policy makers

Policymakers are responsible for creating the regulatory framework that governs the use of HIoT technologies. Their role is crucial in addressing legal and ethical challenges while promoting innovation.

3.4.1 Challenges

Navigating the complex landscape of healthcare regulations and compliance standards is a significant challenge for HIoT adoption. Ensuring that HIoT devices and systems meet regulatory requirements is essential to avoid legal repercussions and ensure patient safety. Data privacy and security are also critical concerns that must be addressed through robust regulatory measures.

3.4.2 Opportunities

Policymakers have the opportunity to foster innovation in HIoT through supportive policies and regulations. By promoting standards that ensure interoperability and security, they can facilitate the widespread adoption of HIoT technologies. Collaborative efforts between policymakers, industry stakeholders, and academia can drive the development of comprehensive solutions that address the technical, clinical, and regulatory aspects of HIoT.

4 Practical challenges in HIoT deployment

The successful deployment of HIoT in healthcare settings involves addressing several practical challenges, such as, security, latency, reliability, efficiency, cost, interoperability, user acceptance, workforce training and other performance-related issues are among the many performance-related issues that the Healthcare Internet of Things (HIoT) presents [64]. The creation and adoption of efficient HIoT systems may be hampered by these issues. In order to successfully handle these issues, an HIoT system should ideally achieve low latency, high dependability, energy-efficient operation, and robust security [65]. In the section that follows, we will examine the unique difficulties that different writers have encountered while using IoT systems, emphasizing the necessity of finding solutions to these problems in order to improve HIoT.

4.1 Cost

The initial investment for HIoT devices and infrastructure can be substantial. Smaller healthcare facilities might struggle with these costs. However, long-term savings from improved efficiency and patient outcomes can offset these expenses. A thorough cost-benefit analysis is essential before implementation.

4.2 Interoperability

HIoT systems must integrate seamlessly with existing healthcare technologies. This requires standardized protocols and interfaces to ensure compatibility between different devices and platforms. Interoperability challenges can hinder data exchange and affect patient care quality.

4.3 User acceptance

Adoption of HIoT technologies depends on the willingness of healthcare professionals and patients to use these systems. Training and education are critical to increase user acceptance. Demonstrating the benefits and ease of use can help mitigate resistance.

4.4 Workforce training

Healthcare providers need to be trained to use HIoT devices effectively. This includes understanding how to operate the devices, interpret the data, and integrate this information into clinical decision-making. Ongoing training programs are necessary to keep pace with technological advancements.

4.5 Latency

Achieving low latency is essential for a number of crucial operations in the Healthcare Internet of Things (HIoT) space, including real-time data processing for emergency response systems, remote patient monitoring, and telemedicine. Low latency is crucial, as it allows medical professionals to act quickly in the event of crises and make timely decisions. However, obtaining low latency in the HIoT may be challenging and often requires specialized hardware and software in addition to improvements to network infrastructure. In the HIoT space, timely data transmission and real-time data analysis provide important problems since any processing and transmission lags might impair the decision-making capacity of healthcare professionals. Achieving low latency in end-to-end transmission and data processing is essential given the time-critical nature of HIoT systems. Using communication solutions with high availability and capacity will help to reduce transmission delays. Machine learning (ML) methods, particularly those related to channel access and routing, can drastically minimize end-to-end delay, delivering an advantage over typical cloud computing models. When combined with AI, fog, and edge computing frameworks, they have been shown to be effective [66]. Deep learning techniques improve the capabilities of big data analytics systems, however, there are still difficulties in obtaining accurate data from enormous datasets. Powerful algorithms that can efficiently exploit and fill in missing data in datasets are urgently needed. The use of computer resources close to the network edge can reduce transmission and processing delays to solve these issues. Fog computing can be included in HIoT to lessen the negative effects of cloud-based latency on emergency response systems. However, issues with reaction time, latency, and energy use still need to be addressed. The Energy-Efficient Internet of Medical Things to Fog Interoperability of Task Scheduling (EEIoMT) framework has been established in this regard [67, 68]. By giving priority to jobs that must be finished in a certain amount of time while controlling energy usage and effectively handling other tasks, this framework improves task scheduling.

4.6 Security

Due to the private and sensitive nature of patient data, security is of utmost importance in the Healthcare Internet of Things (HIoT) industry. H-IoT systems and devices are more susceptible to security breaches and cyberattacks as a result of their increased connection and accessibility. Cybercriminals engaged in identity theft, fraud, and other illegal acts find medical data of great value. Additionally, a security lapse in an H-IoT system might put patients in grave danger by exposing private medical data and perhaps interfering with medical equipment. To protect the confidentiality, integrity, and availability of patient data in HIoT systems, strong security measures are required. They must also guarantee the dependability and safety of medical gadgets. The resource-constrained nature of HIoT devices makes it difficult to deploy robust security mechanisms, which presents another difficulty. The adoption of energy-efficient machine learning or deep learning algorithms that can handle lightweight encryption is necessary since energy efficiency is a major problem [69–71]. Computational complexity is still taken into account in these systems, though. Therefore, using a federated system close to the devices in a fog computing paradigm can improve security while retaining the energy economy. The vulnerability to intermediary attacks and the lack of security knowledge among novice users provide security issues in the context of the Internet of Medical Things (IoMT). A technique for security enhancement called RECC-VC was suggested to lessen these concerns. This approach uses blockchain technology, optimized neural networks, and K-anonymity to give greater precision and security than prior approaches [72]. These developments are essential for protecting the privacy and accuracy of patient data in IoMT systems.

4.7 Platforms for real-time operations

Data gathered from sensors may generally be used to better understand the state of health of a user. In order to extract useful information from the large amount of generated data, it is vital to keep in mind that specialized processing techniques are required. Real-time data handling techniques are required since many existing algorithms have been demonstrated to be ineffective in processing the entire data set. Although deep learning algorithms can handle enormous amounts of real-time data, it is essential to remove various redundant characteristics from the dataset in order to acquire a thorough understanding of the user's health state [73]. This underlines the requirement for sophisticated algorithms that can effectively analyze and make sense of the vast amount of data produced in real time by healthcare sensors.

4.8 Consumption of energy

While implants need sustainable batteries to enable long-term trouble-free operation, wearable healthcare Internet of Things (HIoT) devices can be recharged after a set operational duration. In the literature, a number of techniques have been suggested to overcome the energy restrictions imposed by implants. One such approach is to reduce the amount of outside interference during the implantation process, which helps avoid dangerous side effects during installation [74]. When working with IoT devices, efficient resource management is crucial because of their limited processing, power, and memory. To increase network longevity, these constraints must be overcome [75–77]. There are short-range power solutions that use very little power in the context of fog and edge computing. However, in certain circumstances, availability might be a problem. To be effective, energy harvesting techniques for IoT nodes must adhere to a set of standards. They should be affordable, charge quickly, and most importantly, user-safe. It is essential to address the energy requirements of IoT devices, especially implants, for long-term and reliable operation in healthcare applications.

4.9 Scalability

Although the increasing use of healthcare Internet of Things (HIoT) technology has enormous potential to improve healthcare services, it also comes with a number of significant difficulties. Along with the creation of strong data delivery methods that ensure the integrity of medical data, ensuring the scalability of these systems to handle an increasing number of devices and data sources is crucial. Accurate localization and network coverage are crucial components of routing strategies because healthcare environments are dynamic. Interoperability and excellent traffic management are essential for standardizing devices and processing data efficiently. Addressing mobility challenges, such as those related to sensors on mobile sources such as the human body, is essential to preserve reliable services while users are on the move. Network calibration, particularly in the setting of ever-changing nodes, provides a difficulty. Realizing the full potential of H-IoT in healthcare requires addressing these issues, as shown in [75].

5 Meaning of HIoT in the medical sector

Healthcare Internet of Things (HIoT) is developing as a game-changing solution for the numerous issues that providers confront in the modern healthcare environment. By sending accurate information where and when it is required, H-IoT greatly improves medical intelligence. It starts with simplifying healthcare workflows and removing menial chores, which not only lowers service costs but also allows healthcare personnel to spend more time with patients in a more meaningful way, as noted in [76]. Higher patient happiness and better patient outcome ratings follow, which may raise third-party payments and total income. The medical industry has a significant economic effect, generating over $200 billion in revenue in 2012, according to the IMS Institute [77], and HIoT improves both patient care and corporate success. Additionally, the increased medical intelligence provided by HIoT is vital in lowering the risk of clinical mistakes. Recent developments in biometric sensors and gadgets provide immediate input to healthcare professionals. This implies that medical staff may respond right away rather than having to wait up to two weeks for a patient's reaction, leading to more rapid and efficient care. HIoT provides additional advantages, such as enhanced patient care, when combined with big data analytics. The Internet of Everything (IoE) most critically gives the healthcare industry the crucial knowledge needed to not only treat illnesses but also to preserve general well-being, underscoring its enormous influence on the healthcare environment [78].

6 HIoT applications

H-IoT greatly improves hospital patient visit forecasting in the medical area. Healthcare institutions may forecast patient admission rates and staff availability with extreme accuracy on a daily and hourly basis thanks to the use of big data and time series analysis. These forecasts are made feasible by looking at previous admission data, enabling hospitals to allocate personnel more effectively, shorten wait times for patients, and improve the standard of care for all patients. Electronic Health Records (EHRs) are among the most prevalent and useful H-IoT applications in the healthcare industry. EHRs centralize and digitize patient information, including lab test results, medical histories, and other data, so that authorized healthcare practitioners may easily access them. The conversion of paper-based records to electronic formats provides a number of benefits, including effective data exchange, quick access to a patient's medical history, and automatic reminders for refilled prescriptions and required lab tests. These solutions not only lighten the stress on the office staff but also improve patient care and security.

Another essential feature of H-IoT systems is real-time alerts in the healthcare industry. Clinical decision support systems (CDS) have been used by hospitals for a long time to help medical personnel make quick judgments about patient care. However, putting these mechanisms into place may be expensive. Personal analytics has been a viable strategy in recent years. Patients may now continually check their health data via wearable technology, such as smartwatches. The cloud is then used to store and send this data in a secure manner. The patient's private doctor is informed right away if any worrying changes take place, such as a substantial rise in blood pressure. These in-the-moment notifications allow medical practitioners to act quickly, lowering the dangers of taking untimely action. For instance, inhalers with GPS capability can monitor trends in asthma in both specific patients and on a larger scale. Asthma patients' treatment regimens can be customized with the use of data gathered through clinical decision support. Patient involvement in healthcare is improved with HIoT. People are becoming more health-conscious as a result of adopting smart gadgets to track their vital signs, exercise levels, and daily routines. Heart disease's early warning signs might include conditions like persistent sleeplessness and elevated heart rate. A plethora of self-monitored health data has been produced through the spread of smart gadgets. Medical professionals can access health trends that are identified by HIoT systems once they have uploaded them to the cloud for analysis. The benefits of continuous monitoring for patients include more independence, a considerable drop in hospital visits, and more proactive healthcare management. The use of big data analytics in healthcare is enabling strategic planning that is well-informed. Heat maps, for instance, can show the prevalence of chronic illnesses in a community, assisting healthcare organizations in making data-driven decisions about resource allocation. The availability of real-time data on the performance of medical services, particularly in intensive care units, is essential for strategic planning. With the use of this information, healthcare methods may be enhanced, possibly leading to lower costs and better patient care.

Finding a cancer cure is one of the healthcare industry's most ambitious uses of the HIoT. The Cancer Moonshot initiative makes use of the potential of big data by compiling information from a variety of sources, such as genetics, patient treatment records, and cancer tissue samples kept in biobanks. To find patterns and interactions among cancer proteins, these data are combined and examined. As a consequence, researchers may sketch out therapy regimens and investigate cutting-edge cancer treatment strategies. This method, meanwhile, runs into problems with data integration and interoperability. In the healthcare industry, predictive analytics has become more popular. To evaluate the quality of healthcare service, large-scale research programs gather huge information from millions of individuals. These sophisticated technologies let medical professionals make decisions based on the information at hand, improving patient care and perhaps even averting health problems. Patients with complicated medical histories and numerous chronic diseases benefit most from predictive analytics techniques. They can forecast the likelihood of developing diabetes, high blood pressure, and heart disease, enabling prompt treatments and individualized care programs. As mentioned in [86, 87], the proposed approach has been compared to other current systems in terms of accuracy for predicting heart disease. These systems include ANN, LR, KNN, NB, DT, GV + SVM, SVM, MLP, and SVM + MLP. The analytical results showed that the IoT approaches obtained a remarkably high degree of accuracy, indicating its great potential for use in the healthcare industry as illustrated in Fig. 5.

By enabling remote monitoring and consultation, telehealthcare has revolutionized the way healthcare services are provided. Telehealthcare connects patients with healthcare practitioners regardless of location by utilizing mobile technologies, computers, and telecommunications. The development of several technologies, including smartphones and wearable gadgets, has made it possible to monitor patients remotely and provide consultations via online video conferences. Telehealth has considerably decreased the number of hospital admissions, enhanced patient care, and raised service standards. Big data has changed the area of medical imaging.

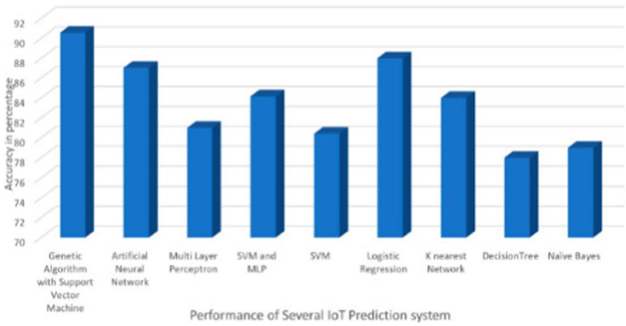

Fig. 5 A number of prediction methods' accuracy in predicting heart disease was evaluated, as is described in references [86, 87]

Advanced algorithms like Carestream and HIoT technologies have made it possible to analyze enormous volumes of medical photos effectively. These algorithms can identify complex patterns in medical pictures, turning them into numerical information that can help with patient care and diagnosis. As it relieves the strain on radiologists who would otherwise have to manually analyze countless intricate medical pictures, this technology has shown to be both time- and money-efficient.

In conclusion, there is no denying the importance of HIoT in the medical industry across a variety of dimensions. It boosts patient interaction, supports predictive analytics, changes telehealth care, revolutionizes medical imaging, improves predictive capacities, and allows data-driven strategic planning. These applications reflect important developments in medical data management and healthcare, resulting in systems that are more effective, patient-centered, and data-informed [79–89]. Results pertaining to the healthcare sector and solutions utilizing the Healthcare Internet of Things (HIoT) are summarized in Table 1. This table probably includes statistics or information that demonstrates how H-IoT is used in the healthcare industry.

7 Critical analysis

According to the various cryptographic platforms taken into consideration, the graph shown in Fig. 6 provides a breakdown of the number of research publications included in the review. The advantages of applying blockchain technology in the healthcare industry have frequently been emphasized in these articles. It is important to keep in mind, though, that these studies usually fall short of addressing the difficulties involved, such as problems with scalability, interoperability, and legal compliance. Few academics have undertaken thorough analyses of the constraints of edge computing, despite the fact that many have looked into the prospective role it may play in healthcare. Although edge computing may considerably speed up and improve the efficiency of data processing, it requires a sizable amount of infrastructure and computer capacity, which could be problematic for smaller healthcare companies. Additionally, while some academics have offered a general overview of the possible uses of AI in healthcare, they have not gone to great lengths to examine its drawbacks and ethical issues.

8 Conclusions

IoT device adoption in healthcare must address security and privacy problems head-on if it is to be widely adopted. There have been several techniques investigated to increase security, however, they all have drawbacks. With a focus on big data, blockchain, machine learning, deep learning, edge computing, and software-defined networks, this article has offered a thorough analysis of safe data access utilizing cryptographic platforms within the context of Healthcare IoT (HIoT). Both prospective uses and the difficulties posed by HIoT have been carefully investigated. In addition, the paper discusses current developments and HIoT's potential for the future, opening the door for cutting-edge solutions that can enhance patient care and healthcare

Table 1 The difficulties facing the healthcare sector today that may be resolved by IoT technology were investigated through a survey

Difficulties	IoT Services	References
not enough modern hospitals	The IoT contribute to the creation of smart hospitals through dedicated networks and automation. These IoT devices incorporate sensors to offer valuable patient data. Smart hospitals digitize information and effectively minimize patient waiting times. Furthermore, data analysis plays a vital role in daily hospital operations and in enhancing the care provided to patients, including those with COVID-19	[90, 91]
Patient data storage for COVID-19	The IoT device efficiently transmits, stores, and analyzes patient data to improve future treatments, facilitating recovery monitoring and enhancing awareness of COVID-19 causes	[92, 93]
Providing treatment to patients with COVID-19	To efficiently monitor COVID-19 patients, a variety of clinical tools, including pumps, weights, and nebulizers, are used. Environmental factors like humidity, temperature, and others may be monitored and controlled by IoT devices. IoT devices have real-time location capabilities that are used to treat COVID-19 patients	[94, 95]
Insufficient precision in decision-making	IoT devices play a valuable role in enhancing the precision of decision-making by predicting and continuously monitoring patient data. Effective communication is essential for accurate surgeries, and IoT technology records and analyzes data, contributing to high-quality decision-making	[96, 97]
Inadequate patient monitoring	IoT devices provide continuous monitoring of hygiene and infection levels in healthcare facilities and support areas. They also ensure accurate medication and protein administration to patients. Furthermore, these devices track the daily progress of a patient's condition. The cloud plays a crucial role in maintaining patient data privacy	[98, 99]

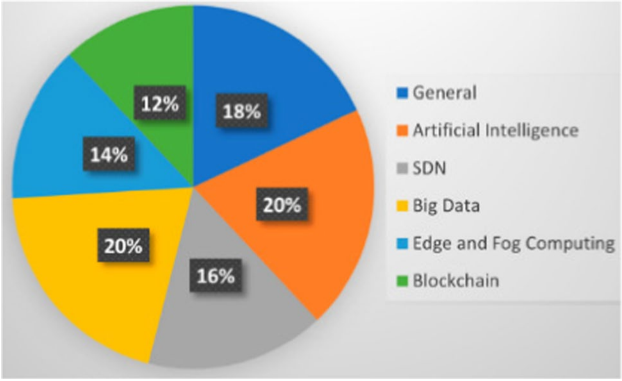

Fig. 6 visualization of the many IoT features that are covered in this study

outcomes. The suggested solutions provide a means of addressing the issues raised while encouraging the wider use of dependable H-IoT systems. As a result, the study emphasizes the significance of legislative initiatives and regulations that provide frameworks for cooperation between governmental entities and the academic community in the field of HIoT. It is crucial to protect user security and privacy while using IoT in healthcare, and this study promotes investigating cryptographic systems like H-IoT to do so. It explores the difficulties and possible uses of HIoT, providing advice on how to create trustworthy systems and serving as a useful resource for academics with an interest in the area.

9 Future research directions

In this paper, we have explored the trends, prospects, challenges, and security aspects of the Healthcare Internet of Things (HIoT). Our analysis highlights the significant potential of HIoT to transform healthcare delivery through enhanced patient monitoring, improved diagnostics, and personalized treatment. The integration of cryptographic platforms within HIoT systems is crucial to ensuring secure data access and protecting patient privacy. Despite the promising advancements, several challenges remain, particularly in the areas of data privacy, interoperability, and infrastructure scalability.

Therefore, the rapid evolution of HIoT presents numerous avenues for future research. This section explores emerging technologies, novel applications, unresolved challenges, and opportunities for collaboration among academia, industry, and policymakers to advance HIoT in healthcare.

9.1 Emerging technologies

- *5 G and Beyond* The deployment of 5 G networks will significantly enhance the connectivity, speed, and reliability of HIoT devices. Future research can focus on integrating 5 G with HIoT to support real-time data transmission, remote surgeries, and augmented reality applications in healthcare.

- *Artificial Intelligence (AI) and Machine Learning (ML)* AI and ML can revolutionize HIoT by providing predictive analytics, personalized treatment plans, and intelligent monitoring systems. Research can delve into developing sophisticated algorithms that can analyze vast amounts of healthcare data generated by IoT devices to improve patient outcomes.
- *Blockchain Technology* Blockchain offers a secure and transparent method for managing healthcare data. Future studies can explore the implementation of blockchain to ensure data integrity, enhance privacy, and facilitate secure data sharing among healthcare providers.
- *Quantum Computing* As quantum computing evolves, it holds the potential to solve complex computational problems much faster than classical computers. Research can investigate its application in processing large datasets generated by HIoT devices, leading to more accurate diagnostics and treatment plans.

9.2 Novel applications

- *Remote Patient Monitoring* With the increasing prevalence of chronic diseases, remote patient monitoring through HIoT devices can provide continuous health tracking and timely medical interventions. Research can focus on developing more advanced and user-friendly monitoring systems.
- *Smart Hospitals* Integrating HIoT in smart hospitals can optimize resource management, enhance patient care, and improve operational efficiency. Future research can explore innovative solutions for automating hospital processes and integrating various IoT devices within the hospital infrastructure.
- *Telemedicine* The COVID-19 pandemic has accelerated the adoption of telemedicine. Future research can investigate the integration of HIoT with telemedicine platforms to provide comprehensive remote care, including real-time health monitoring and virtual consultations.

9.3 Unresolved challenges

- *Data Privacy and Security* Ensuring the privacy and security of healthcare data remains a critical challenge. Research can focus on developing robust encryption methods, secure authentication protocols, and comprehensive cybersecurity frameworks to protect sensitive health information.
- *Interoperability* The lack of standardization among HIoT devices hinders seamless data exchange and integration. Future studies can address the development of universal standards and protocols to ensure interoperability and compatibility among diverse HIoT systems.
- *Scalability and Infrastructure* As the number of connected devices grows, so does the need for scalable infrastructure. Research can explore solutions for managing large-scale HIoT deployments, including efficient data storage, processing, and network management.

9.4 Opportunities for collaboration

- *Academia-Industry Partnerships* Collaborative efforts between academic institutions and industry can drive innovation in HIoT. Academia can provide cutting-edge research and theoretical insights, while industry can offer practical applications and resources for large-scale implementation.
- *Policy and Regulation* Policymakers play a crucial role in shaping the future of HIoT. Research can inform the development of regulations and policies that promote innovation while ensuring patient safety, data privacy, and ethical standards.
- *International Collaboration* Global collaboration can accelerate the advancement of HIoT by sharing knowledge, resources, and best practices. Research can focus on establishing international frameworks for cooperation and addressing global health challenges through HIoT solutions.

In conclusion, the future of HIoT in healthcare is promising, with significant potential to transform patient care and improve healthcare outcomes. By addressing emerging technologies, novel applications, unresolved challenges, and fostering collaboration, the HIoT ecosystem can advance towards a more secure, efficient, and patient-centered healthcare system.

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